

ClaimBot
Insurance Claims Automation Platform
Use Case Scenario Document

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1. Executive Summary

Project Name

ClaimBot - Insurance Claims Automation Platform

Overview

ClaimBot is an AI-powered conversational agent designed to automate the First Notice of Loss (FNOL) claims intake process for automobile insurance. The system provides a guided, empathetic conversational experience that collects comprehensive incident information, automatically detects claim scenarios, performs intelligent triage routing, and seamlessly escalates complex cases to human specialists. Built on a state machine architecture with LangGraph orchestration, ClaimBot supports multiple LLM providers (OpenAI, AWS Bedrock, Ollama) and integrates document OCR/vision capabilities for evidence extraction and verification.

Key Features

Feature	Description
Conversational FNOL	11-state guided conversation flow for structured data collection
Scenario Playbooks	20+ specialized playbooks for different incident types (collision, theft, weather)
Intelligent Triage	Deterministic scoring engine routes claims to STP, Adjuster, SIU, or Emergency
Document OCR	Vision-capable LLMs extract data from photos and documents
Human Escalation	Celest specialist portal with live chat takeover capability
Multi-LLM Support	OpenAI GPT-4o, AWS Bedrock Claude, and Ollama support

Target Users

Policyholders filing auto insurance claims after accidents, theft, or damage incidents

Insurance claims specialists (Celest agents) handling escalated cases

Insurance administrators managing system configuration and monitoring operations

Core Value Proposition

ClaimBot reduces claim intake time from 20-30 minutes to under 10 minutes while improving data quality through structured collection and intelligent validation. The system provides 24/7 availability for claim filing, reduces operational costs through automation of routine claims, and ensures consistent customer experience with empathetic, guided interactions. Complex cases are automatically identified and routed to human specialists with complete context preservation.

2. Problem Statement

Filing an auto insurance claim after an accident is a stressful, time-consuming process that typically requires customers to navigate phone queues, repeat information multiple times, and wait for business hours to report incidents. Traditional claim intake processes are inefficient, prone to human error, and fail to capture comprehensive incident details consistently.

Who Experiences It

Insurance policyholders who have experienced auto accidents, vehicle theft, weather damage, or other covered incidents. These customers are often in stressful situations requiring immediate assistance but face long wait times and complex procedures. Insurance companies also suffer from inconsistent data collection, high operational costs, and delayed claim processing.

Why It Matters

- Customer satisfaction drops significantly when claim filing is difficult during already stressful situations
- Incomplete or inconsistent claim data leads to processing delays and increased claim handling costs
- Limited business hours restrict when customers can file claims, causing frustration and potential coverage disputes
- Manual data entry and validation consumes significant adjuster time that could be spent on complex cases
- Fraud detection becomes harder without structured, consistent data collection
- Limitations of Existing Approaches
- Phone-based intake requires wait times and business hours availability
- Web forms are impersonal and fail to adapt to specific incident types
- Generic chatbots lack domain expertise and cannot handle complex scenarios
- Manual processes result in inconsistent data quality across claims
- Existing systems do not integrate document verification or intelligent triage

3. Target User Persona

Primary User: Policyholder (John)

Role: Auto insurance policyholder who has just experienced a vehicle incident (collision, theft, or damage)

Scenario: Just involved in a two-vehicle collision at an intersection. Safe but stressed, he needs to file a claim quickly from his smartphone.

Goals

- File an insurance claim quickly and efficiently after an incident
- Understand what information is needed and what to expect next
- Receive confirmation that the claim has been submitted successfully

- Get connected to a human specialist if the situation is complex or urgent

Constraints

- May be filing from an accident scene with limited time and attention
- May not have all documents immediately available
- Unfamiliar with insurance terminology and claim processes
- Emotionally stressed and seeking reassurance

Secondary User: Claims Specialist (Celest Agent)

Role: Insurance claims specialist who handles escalated cases requiring human judgment

Scenario: Handles escalated cases that require human judgment - injuries, suspected fraud, high-value claims, and customer requests for specialist support.

Goals

- Review escalated cases efficiently with complete context
- Communicate with customers via live chat when needed
- Approve, deny, or request additional information on claims
- Maintain audit trail of all actions taken

4. Current Workflow (Before the Agent)

Step-by-Step Traditional Process

Step 1: Incident Occurs: Customer experiences auto accident, theft, or damage. Often in a stressful state, possibly at the accident scene.

Step 2: Contact Insurance Company: Customer must wait until business hours, call the claims hotline, and navigate IVR menus. Average wait time: 10-15 minutes.

Step 3: Speak with Claims Representative: Customer explains the incident verbally while representative manually enters information into the system. Representative asks standard questions that may not be relevant to the specific incident type.

Step 4: Provide Documentation: Customer must separately email or fax supporting documents (police reports, photos, repair estimates). No real-time verification or feedback.

Step 5: Await Processing: Claim is queued for processing. Customer receives little visibility into status. May receive follow-up calls requesting additional information.

Step 6: Claim Decision: After days or weeks, customer receives claim decision. If information was missing, process restarts.

Pain Points

- Long wait times during phone calls, especially after major weather events
- Business hours restrictions prevent immediate filing

- Information must be repeated multiple times across different touchpoints
- No visibility into what specific information is needed for the incident type
- Document submission is disconnected from initial filing
- No real-time validation or feedback on submitted information
- Complex cases are not identified early, leading to delays

Inefficiencies

- Manual data entry by claims representatives introduces errors
- Generic question sets do not adapt to specific incident scenarios
- No automated fraud detection or triage at intake
- Specialist time wasted on routine claims that could be auto-processed
- Document verification happens late in the process, causing rework

5. Use Case Scenario (Primary Scenario)

Scenario: Two-Vehicle Collision Claim Filing

Trigger: Policyholder John has been involved in a two-vehicle collision at an intersection. He is safe and has exchanged information with the other driver. He opens the insurance company's customer portal on his smartphone and clicks "File a Claim".

Agent Processing Steps

	State	Action
1	SAFETY_CHECK	Verify all parties are safe, check for emergencies
2	IDENTITY_MATCH	Verify policyholder identity, retrieve policy details
3	INCIDENT_CORE	Collect date, time, location, basic description
4	LOSS_MODULE	Classify incident type, activate relevant playbooks
5	VEHICLE_DRIVER	Confirm involved vehicle, collect damage details
6	THIRD_PARTIES	Collect other driver/vehicle information
7	INJURIES	Assess injuries to all parties
8	DAMAGE_EVIDENCE	Request photo uploads, analyze with OCR/Vision
9	TRIAGE	Evaluate claim against rules, determine routing
10	CLAIM_CREATE	Present summary, create claim on confirmation
11	NEXT_STEPS	Provide claim number, explain next steps

Tools and APIs Used

Service	Function
LangGraph	State machine orchestration for 11-state conversation flow
Intent Service	Classifies user responses (confirm, provide info, ask question)
Playbook Registry	Detects two_vehicle collision scenario, injects specific questions
OCR/Vision Service	Extracts damage information from uploaded photos
Triage Engine	Scores claim (45 points) - routes to STP (below 200 threshold)
Policy Validation	Verifies active coverage for the involved vehicle
Session Store	Redis-based state persistence across conversation turns
Database (PostgreSQL)	Persists claim data and audit trail

Message Processing Flow:

1. User message received via WebSocket or REST API
2. Session state retrieved from Redis cache
3. Current state handler invoked with user input
4. LLM extracts structured fields from natural language
5. Playbook registry injects scenario-specific questions
6. Response generated with next question
7. State updated and persisted to Redis + PostgreSQL
8. On completion, ClaimDraft converts to Claim record

System Decisions

Decision Point	Outcome
Playbook Detection	"hit by another car" triggers two_vehicle playbook (0.9 confidence)
Triage Score	No injuries (0) + drivable (+0) + police report (+15) + two vehicles (+30) = 45 pts
Routing	Score 45 < 200 threshold → Straight-Through Processing (STP)
Evidence Check	4 photos uploaded → meets minimum evidence requirement

Output Generated

- Claim record with unique claim number (e.g., CLM-2024-00123)

- Structured claim data including all collected fields
- Uploaded documents linked to claim record
- Audit trail of conversation and state transitions
- Triage result with routing decision and scoring breakdown

Resulting Outcome

John receives his claim number within 8 minutes of starting the process. He can track his claim status on his dashboard. An adjuster is automatically assigned and will contact him within 24-48 hours. All relevant information has been captured in a structured format, eliminating the need for follow-up calls.

6. Functional Capabilities

Conversational FNOL Intake

The system provides a guided conversational experience through an 11-state flow: Safety Check, Identity Match, Incident Core, Loss Module, Vehicle/Driver, Third Parties, Injuries, Damage/Evidence, Triage, and Claim Creation. Each state collects specific information while maintaining conversational context and allowing natural language responses.

Scenario-Specific Playbooks

The system includes 20+ specialized playbooks that automatically activate based on incident characteristics:

- Collision Playbooks: Two-vehicle, single-vehicle, multi-vehicle (3+), hit-and-run, uninsured motorist, parking lot, animal strike
- Weather Playbooks: Hail damage, flood damage, wind/tree damage
- Theft Playbooks: Vehicle theft, attempted theft
- Other Playbooks: Vandalism, glass-only, fire, towing, commercial/rideshare, rental vehicle, out-of-state incident, injury, severe injury, police/DUI involvement

Intelligent Triage Engine

A deterministic scoring engine evaluates each claim against configurable rules to determine routing:

- STP (Straight-Through Processing): Score below 200 points, automated processing
- Adjuster Review: Score 200+ points or injury present, assigned to adjuster
- SIU Review: DUI suspected or fraud indicators, Special Investigations Unit
- Emergency: Severe injury or fatality, immediate priority handling

Document OCR and Verification

Vision-capable LLM models extract structured data from uploaded documents, including driver's licenses, police reports, repair estimates, and damage photos. Cross-document verification identifies discrepancies in dates, locations, amounts, and party information.

Human-in-the-Loop Escalation (Celest)

When claims require human judgment, they are escalated to the Celest specialist portal. Features include:

- Escalation queue with SLA tracking and priority indicators
- Complete conversation history and collected data visibility
- Live chat takeover capability for real-time customer communication
- Case locking to prevent concurrent agent conflicts
- Approve, deny, or request additional information actions
- Internal notes and audit trail

Multi-LLM Provider Support

The system supports multiple LLM providers configurable via the admin dashboard:

- OpenAI GPT-4o and GPT-4o-mini for text and vision tasks
- AWS Bedrock with Claude 3 Sonnet for enterprise deployments
- Ollama for local/on-premise deployments (Llama3, LLaVA for vision)

Admin Configuration Dashboard

- LLM provider selection and API key configuration
- Flow rules and auto-approval thresholds
- Document flow sequences by product line and incident type
- Metrics dashboard with claims statistics and escalation rates
- Session monitoring and transcript review
- Comprehensive Audit log access

Authentication and Authorization

The system implements role-based access control with three primary roles:

- Customer: Can file claims, view their policies and claim status, and upload documents
- Celest (Specialist): Can view escalation queue, take over chats, approve/deny claims
- Admin: Full system configuration access, metrics, audit logs, and user management

7. Technical Architecture

System Components

Layer	Technologies
Backend	Python 3.11, FastAPI, LangGraph, SQLAlchemy, Redis
Frontend	React 18, TypeScript, Vite
Database	PostgreSQL 15 with Alembic migrations

Cache	Redis for session state management
AI/ML	LangChain, LangGraph, OpenAI/Bedrock/Ollama
Observability	LangFuse for LLM tracing (optional)

Frontend Application

- Customer Portal: FNOL chat widget, policy dashboard, claims tracking
- Admin Dashboard: System configuration, metrics, transcripts
- Celest Portal: Escalation queue, case detail view, live chat

Agent Design Pattern

ClaimBot uses a state machine architecture implemented with LangGraph. The conversation flow is divided into discrete states, each responsible for collecting specific information. State handlers process user input, extract relevant fields, and determine the next state based on collected data and business rules.

Key Design Patterns:

- State Machine: 11 sequential states with defined transitions
- Playbook Pattern: Extensible scenario-specific logic modules
- Service Layer: Encapsulated business logic (Intent, Extraction, Triage)
- Repository Pattern: Data access abstraction via SQLAlchemy

Model Usage

- Intent Classification: Pattern matching with optional LLM fallback for ambiguous inputs
- Field Extraction: LLM extracts structured data from natural language responses
- Document OCR: Vision-capable models (GPT-4o, Claude 3, LLaVA) extract entities from images
- Response Generation: LLM generates empathetic, contextual responses for each state

Database Schema

Key database models include:

User: Authentication, roles, profile information

Policy: Insurance policies with coverage details

Claim: Filed claims with status, metadata, timeline

ClaimDraft: In-progress FNOL data before submission

Document: Uploaded files with OCR extraction results

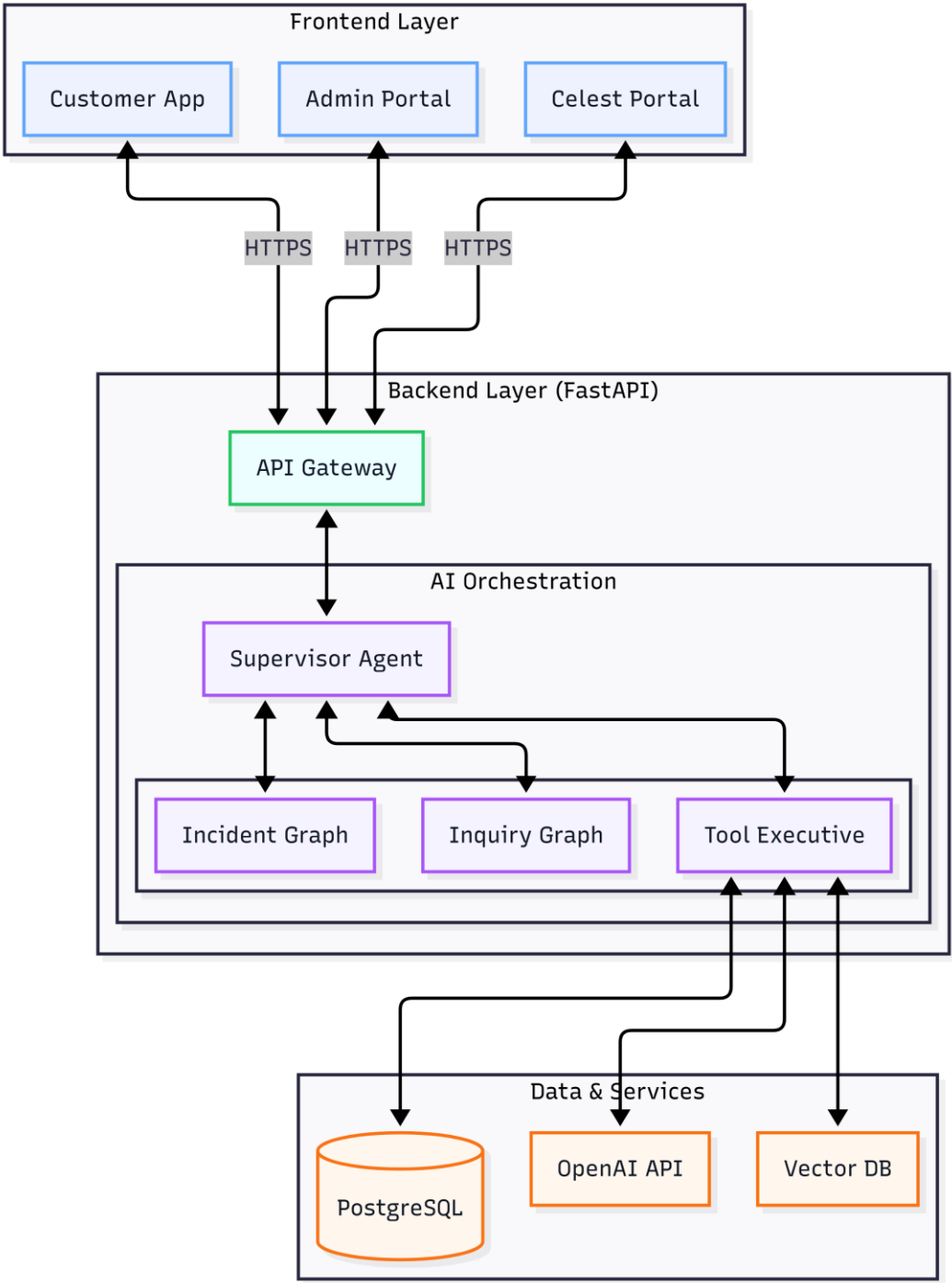
Case: Escalated cases for specialist review

Session: Conversation sessions with message history

SystemSettings: Admin-configurable system parameters

AuditLog: Comprehensive audit trail

High-Level Architecture



8. Edge Cases and Limitations

Known Technical Limitations

- OCR accuracy depends on image quality; poor lighting or blurry photos may result in extraction failures
- LLM responses may occasionally misinterpret ambiguous user input despite pattern matching safeguards
- Large document uploads (>10MB) are rejected to prevent system overload

Scalability Concerns

- Redis memory scaling required for peak concurrent sessions (e.g., major storms)
- LLM API rate limits may constrain throughput during catastrophic events
- PostgreSQL connection pooling needs tuning for high concurrent load

Model Constraints

- Context window limits may truncate very long conversation histories
- Vision models may not accurately assess damage severity from photos
- Intent classification may fail for non-English inputs or heavy dialect
- Extraction accuracy varies by LLM provider and model version

Failure Scenarios

- LLM API unavailability triggers automatic escalation to human specialist
- Session timeout after 15 minutes of inactivity prompts user to continue or end
- Document upload failures allow retry or skip with manual follow-up
- Database connection failures are handled with retry logic and graceful degradation

Implementation Assumptions

- Users have valid insurance policies that can be verified at intake
- Auto insurance claims are the primary product line (architecture supports extension)
- Users can upload photos from their devices
- External policy administration systems would integrate via API (currently simulated)

9. Impact and Value

Key Performance Metrics

Metric	Before	After	Improvement
Claim Intake Time	20-30 minutes	<10 minutes	60-70% reduction
Phone Wait Time	10-15 minutes	0 minutes	Eliminated
24/7 Availability	No (business hours)	Yes	100% coverage
Data Completeness	~70%	~95%	25% improvement
STP Rate	~30%	60-70%	2x improvement

Benefits by Stakeholder

Stakeholder	Benefits
Customers	Faster filing, 24/7 access, clear guidance, immediate confirmation
Specialists	Complete context, structured data, live chat capability, audit trail
Company	Reduced costs, consistent data, faster cycle time, better fraud detection

Automation Level

Based on the triage engine configuration, approximately 60-70% of claims (simple collisions without injuries, weather damage with documentation) can proceed through Straight-Through Processing without human intervention. The remaining 30-40% are automatically escalated with complete context, allowing specialists to make decisions faster.

10. Future Improvements

The following enhancements are realistic extensions based on the current architecture:

Near-Term Enhancements

- Multi-language support with translation layer for non-English speakers
- Voice interface integration for hands-free claim filing from accident scenes
- Push notifications for claim status updates and document requests
- Integration with repair shop networks for direct estimate submission
- Enhanced fraud detection rules based on historical claim patterns

Medium-Term Extensions

- Additional product lines (homeowners, renters, commercial auto)
- AI-powered damage severity estimation from photos
- Automated repair shop assignment based on location and specialization
- Customer sentiment analysis for proactive escalation
- Integration with telematics data for accident reconstruction

Long-Term Vision

- Predictive claim outcome modeling for settlement optimization
- Automated subrogation identification and pursuit
- Real-time weather event detection for proactive customer outreach
- Document verification and fraud prevention
- Full claims lifecycle automation from FNOL through payment

Technical Improvements

- GraphQL API for flexible frontend data fetching
- Real-time analytics dashboard with streaming metrics
- A/B testing framework for conversation flow optimization
- Enhanced LangFuse integration for conversation analytics and optimization

11. Appendix A: State Machine Reference

State	Type	Next States	Purpose
SAFETY_CHECK	Initial	IDENTITY_MATCH, HANDOFF	Verify safety, check emergencies
IDENTITY_MATCH	Auth	INCIDENT_CORE, HANDOFF	Verify identity, retrieve policy
INCIDENT_CORE	Collection	LOSS_MODULE	Date, time, location, description
LOSS_MODULE	Classification	VEHICLE_DRIVER	Classify type, activate playbooks
VEHICLE_DRIVER	Collection	THIRD_PARTIES	Vehicle details, driver info
THIRD_PARTIES	Collection	INJURIES	Other party information
INJURIES	Assessment	DAMAGE_EVIDENCE, HANDOFF	Injury assessment for all parties
DAMAGE_EVIDENCE	Collection	TRIAGE	Photos, documents, OCR extraction
TRIAGE	Decision	CLAIM_CREATE, HANDOFF	Score calculation, routing decision
CLAIM_CREATE	Creation	NEXT_STEPS, HANDOFF	Summary review, claim creation
NEXT_STEPS	Terminal	-	Claim number, next steps explained
HANDOFF_ESCALATION	Terminal	-	Escalated to human specialist

12. Appendix B: Triage Rules Reference

Hard Rules (Override Score)

Rule ID	Condition	Route
severe_injury	Severe/fatal injury	EMERGENCY
dui_suspected	DUI/DWI suspected	SIU_REVIEW
injury_any	Any injury reported	ADJUSTER

Scoring Rules

Rule ID	Condition	Points
theft	Vehicle theft reported	+100
multi_vehicle	3+ vehicles involved	+80
vehicle_not_drivable	Insured vehicle not drivable	+60
hit_and_run	Hit-and-run incident	+50
commercial_use	Commercial/rideshare use	+50
property_damage	Third-party property damage	+40
cross_border	Out-of-state incident	+40
guest_mode	No policy matched	+30
tow_required	Towing required	+20
police_involved	Police contacted	+15
glass_only	Glass-only with photo	-50

Routing Threshold: Score $\geq$ 200 points routes to ADJUSTER; below 200 routes to STP.

13. Appendix C: Sample Conversation

Bot: Hello! I'm ClaimBot, and I'm here to help you file your auto insurance claim. Before we begin, I need to confirm - are you and everyone involved in the incident safe? Is there any emergency that needs immediate attention?

User: Yes, everyone is safe. No emergencies.

Bot: I'm glad to hear everyone is safe. I've verified your identity and found your policy ending in ...4523 for a 2022 Honda Accord. Is this the vehicle involved in the incident?

User: Yes, that's the one.

Bot: Thank you. Can you tell me when and where this incident occurred?

User: It happened today around 2:30pm at the intersection of Main St and 5th Ave.

Bot: I've recorded the incident as today at approximately 2:30 PM at Main St and 5th Ave. Can you briefly describe what happened?

User: I was hit by another car that ran a red light. It was a two-car collision.

Bot: I understand - a two-vehicle collision where the other driver ran a red light. Is your vehicle drivable, or does it need to be towed?

User: It's drivable, just has damage to the front passenger side.

Bot: Thank you. Were you able to get the other driver's information - their name, phone number, and insurance details?

User: Yes, his name is John Smith, phone 555-1234, he has State Insurance policy number SI-789456.

Bot: I've recorded the other driver's information. Were there any injuries to you, your passengers, or anyone in the other vehicle?

User: No, no injuries for anyone.

Bot: That's good to hear. Could you please upload some photos of the damage to your vehicle? This will help us process your claim faster.

User: [Uploads 4 photos]

Bot: Thank you! I've received 4 photos showing damage to the front passenger side. Based on the information you've provided, I've prepared a summary of your claim. Please review and confirm the details are accurate...

14. Glossary

Term	Definition
FNOL	First Notice of Loss - The initial report of an insurance claim
STP	Straight-Through Processing - Automated claim processing without human intervention
SIU	Special Investigations Unit - Team handling potential fraud cases
Celest	Claims Specialist portal for handling escalated cases
Playbook	Scenario-specific logic module for different incident types
Triage	Process of evaluating and routing claims based on complexity
OCR	Optical Character Recognition - Extracting text from images
LangGraph	Framework for building stateful AI agent applications
LLM	Large Language Model - AI models like GPT-4, Claude